

QUICK REVISION DECK

GitHub Actions & CI/CD

Revision Notes — MLOps Bootcamp, Video 8

Concepts • YAML anatomy • ready-to-reuse templates • interview recall

Code



CI Test



Build



Deploy

What's Inside This Deck

1

Concepts

CI/CD, GitHub Actions, key terms, dev workflow, deploy environments

2

YAML Anatomy

The workflow file decoded field-by-field, so you can read any pipeline

3

Templates

Pytest CI workflow + Docker build & push workflow — copy straight into a project

4

Interview-Ready

Common errors, best practices, and quick-fire Q&A recall

Git, GitHub & GitHub Actions

Git

Distributed version control — track files, branch, merge, resolve conflicts.

GitHub

Remote code repository for collaboration, built on top of Git.

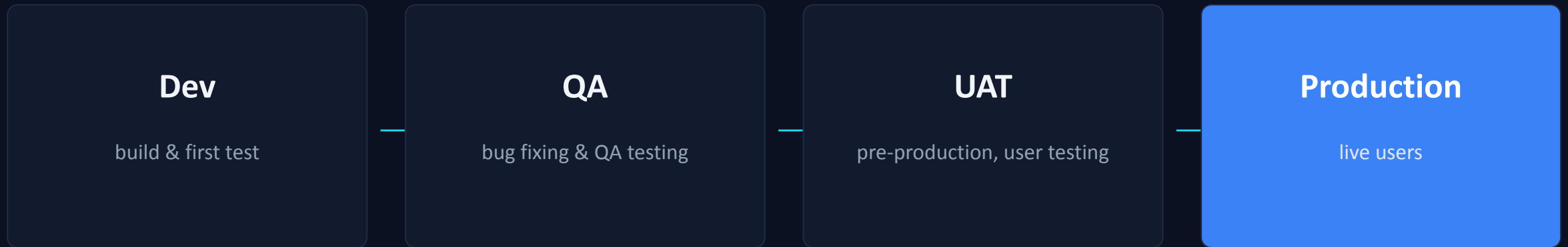
GitHub Actions

Built-in automation engine. Runs workflows triggered by events (push, PR, schedule).

CI vs CD

CI = auto build + test on every merge. CD = auto deploy tested code across environments.

Deployment Environments (CD Path)



Each arrow is an automated deployment step, triggered once the previous stage passes — no manual copy-paste to servers.

The Standard Feature-to-Production Flow



Benefits: fewer errors · faster releases · continuous feedback · better collaboration

Workflow Field Cheat Sheet

Field	Meaning
<code>name</code>	Workflow display name shown in the Actions UI
<code>on</code>	Trigger event(s): push, pull_request, schedule...
<code>branches</code>	Filters which branch(es) trigger the workflow
<code>jobs</code>	Group(s) of work; each job runs in its own container
<code>runs-on</code>	Runner OS, e.g. ubuntu-latest
<code>needs</code>	Job dependency — waits for another job to succeed
<code>steps</code>	Ordered tasks inside a job
<code>uses</code>	Runs a pre-built Action, e.g. actions/checkout@v3
<code>with</code>	Input parameters passed to an action
<code>run</code>	Executes a raw shell command
<code>\${{ secrets.X }}</code>	Securely injects a GitHub encrypted secret

Template 1 — Pytest CI Workflow

Runs on every push/PR to main: checkout → setup Python → install deps → run tests.

```
name: Python CI
on:
  push: { branches: [main] }
  pull_request: { branches: [main] }

jobs:
  test:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: actions/setup-python@v4
        with: { python-version: '3.9' }
      - run: pip install -r requirements.txt
      - run: pytest
```

Template 2 — Docker Build & Push (CI/CD)

Job 2 only runs if Job 1 (tests) passes — that's what "needs" does.

```
jobs:
  build-and-test:
    runs-on: ubuntu-latest
    steps: [checkout, setup-python, install, run: pytest]

  build-and-publish:
    needs: build-and-test
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v3
      - uses: docker/login-action@v2
        with:
          username: ${ secrets.DOCKER_USERNAME }
          password: ${ secrets.DOCKER_PASSWORD }
      - uses: docker/build-push-action@v4
        with: { context: ., file: ./Dockerfile, push: true,
              tags: ${ secrets.DOCKER_USERNAME }/app:latest }
```

Docker Hub Secrets — Setup Steps

1

Docker Hub

Account Settings → Personal Access Tokens → Generate
(Read / Write / Delete).

2

GitHub Repo

Settings → Secrets and variables → Actions → add
DOCKER_USERNAME & DOCKER_PASSWORD (paste the
token).

Never hardcode credentials in YAML — always reference via `${{ secrets.NAME }}`

Common Error, Fix & Habits

Error: failed to read dockerfile: open dockerfile: no such file

Cause: build-push-action couldn't locate the Dockerfile path.

Fix: add file: ./Dockerfile under the with: block.

Best Practices

- Keep venv/ in .gitignore — never commit it
- Pin action versions (@v3, @v4) for reproducibility
- Split CI (test) and CD (deploy) into separate jobs with needs
- Verify locally: docker pull + docker run -p 5000:5000

Quick-Fire Q&A

Q CI vs CD?

CI = auto build/test on merge. CD = auto deploy the tested build across environments (Dev → QA → UAT → Prod).

Q Why use "needs"?

Enforces job order — deployment only runs after tests pass, preventing broken releases.

Q Why GitHub Secrets, not plain text?

Encrypted storage — avoids leaking credentials in source code or logs.

Q What triggers a workflow?

Events under on: — push, pull_request, schedule, manual dispatch, etc.

One-Page Glossary

Term	Definition
Workflow	Automated YAML pipeline defined in <code>.github/workflows/*.yml</code>
Job	A set of steps that run on one runner (container)
Step	Single task inside a job — either uses an action or runs a command
Runner	The VM/container executing the job (e.g. ubuntu-latest)
Action	Reusable pre-built unit of automation (e.g. checkout, setup-python)
Secret	Encrypted credential stored in repo settings



Revision Complete

Reuse the two YAML templates directly in your ML projects for automated testing + Docker deployment.